Cap 4 - Exp. Algébricas

Exemplos

Ex 4.1 a) $5x + 5y = \boxed{5(x+y)}$

b) $3a^3x - 3a^3y - 6a^3 = \boxed{3a^3(x - y - 2)}$

c) $3x + 3y + mx + my = \boxed{(3+m)(x+y)}$

d) $x^3 + x^2 + x + 1 = \boxed{x^3\left(1 + \frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^3}\right)}$

e) $x^2 - 25 = x^2 - 5^2 = \boxed{(x+5)\,.\,(x-5)}$

f) $x^2 - 4x + 4 = \boxed{(x-2)^2}$

02

g) $x^2 + 20x + 100 = \boxed{(x+10)^2}$

h) $x^4 - 22x^2 + 121 = \boxed{(x^2-11)^2}$

i) $y^8 + 4y^4 + 4 = \boxed{(y^4+2)^2}$

Ex 4.2 $\dfrac{3x+3y}{x^2-y^2} = \dfrac{3(x+y)}{(x+y)\cdot(x-y)} = \boxed{\dfrac{3}{x-y}}$

Ex 4.3 $2x^3 - x^2 - x - 10 \;\big|\; x-2$

$-2x^3 + 4x^2 \qquad\qquad 2x^2 + 3x + 5$

$3x^2 - x$

$-3x^2 + 6x$

$5x - 10$

$-5x + 10$

0

Algoritmo: Euclides

Então:

$\boxed{\begin{aligned} Q(x) &= 2x^2 + 3x + 5 \\ R(x) &= 0 \end{aligned}}$

Ex 4.3 Algoritmo: Briot-Ruffini

Temos que: $P(x) = 2x^3 - x^2 - x - 10$
$D(x) = x - 2$

P/ aplicarmos Briot-Ruffini, precisamos de uma raiz de $D(x)$. Aqui, temos que: $x' = 2$

O algoritmo é:

	r	a_1	a_2	a_3	a_4
		b_1			

$b_2 = r \cdot b_1 + a_2$

Então:

2	2	-1	-1	-10
	2	3	5	0

$Q(x)$ $R(x)$

Então, temos que:

$$Q(x) = 2x^2 + 3x + 5$$
$$R(x) = 0$$

Lista de Exercícios

Expressões Algébricas

(01) a) $10x - 10y = \boxed{10(x-y)}$

b) $x^5 - x^3 = \boxed{x^3(x^2-1)}$

c) $ac + bc + ad + bd = \boxed{(a+b)(c+d)}$

d) $8x - 3xy + 8 - 3y =$

$\boxed{(8-3y)(x+1)}$

e) $100 - x^2 = 10^2 - x^2 =$

$\boxed{(10+x).(10-x)}$

f) $4x^2 - 1 = (2x)^2 - 1^2 =$

$\boxed{(2x+1).(2x-1)}$

g) $x^2 - 10x + 25 =$

$\boxed{(x-5)^2}$

02

02 a) $(x-5y)^2 =$ $\boxed{x^2 - 10xy + 25y^2}$

b) $(ab^2+3)^2 =$ $\boxed{a^2b^4 + 6ab^2 + 9}$

c) $\left(\frac{a}{4} - \frac{1}{4}\right)^2 = \frac{a^2}{16} - \frac{a}{8} + \frac{1}{16} =$

$\boxed{\frac{a^2 - 2a + 1}{16}}$

d) $(5x^2-4)(5x^2+4) =$ $\boxed{25x^4 - 16}$

03 a) $\frac{x^2 - x}{x-1} = \frac{x(x-1)}{(x-1)} =$ $\boxed{x}$

b) $\frac{x+2}{x^2+4x+4} = \frac{(x+1)}{(x+2)\cdot(x+2)} =$ $\boxed{\frac{1}{x+2}}$

c) $\frac{a^2-9}{a-3} = \frac{a^2-3^2}{a-3} = \frac{(a+3)\cdot(a-3)}{(a-3)} =$ $\boxed{a+3}$

d) $\frac{x^2-6x+9}{3x-9} = \frac{(x-3)\cdot(x-3)}{3(x-3)} =$ $\boxed{\frac{x-3}{3}}$

(04) a) $x^4 - x^3 + 0x^2 + 2x - 1 \;\big|\; x^2 + x + 1$

$-(x^4 + x^3 + x^2) \qquad x^2 - 2x + 1$

$-2x^3 - x^2 + 2x$

$-(-2x^3 - 2x^2 - 2x)$

$x^2 + 4x - 1$

$-(x^2 + x + 1)$

$3x - 2$

Então: $\boxed{\begin{array}{l} Q(x) = x^2 - 2x + 1 \\ R(x) = 3x - 2 \end{array}}$

b) $x^3 - x^2 - 6x \;\big|\; x + 2$

$-(x^3 + 2x^2) \qquad x^2 - 3x$

$-3x^2 - 6x$

$-(-3x^2 - 6x)$

0

$\boxed{Q(x) = x^2 - 3x}$

c) $x^3 + 2x^2 - 7x + 4 \;\big|\; x^2 - 2x + 1$

$-(x^3 - 2x^2 + x) \qquad x + 4$

$4x^2 - 8x + 4$

$-(4x^2 - 8x + 4)$

0

Então: $\boxed{Q(x) = x + 4}$

04

d) $x^4 - 2x^3 - 15x^2 \;\big|\; x^2 + 3x$

$-x^4 + 3x^3 \qquad\qquad x^2 - 5x$

$-5x^3 - 15x^2$

$-5x^3 - 15x^2$

0

Então:

$$\boxed{Q(x) = x^2 - 5x}$$

05 a) $P(x) = 5x^5 - 14x^3 + 10x^2 - 7x + 7$

$D(x) = x - 2$

2	5	0	-14	10	-7	7
	5	10	6	22	37	81

$\hookrightarrow R(x)$

Então:

$$\boxed{\begin{aligned} Q(x) &= 5x^4 + 10x^3 + 6x^2 + 22x + 37 \\ R(x) &= 81 \end{aligned}}$$

b) $P(x) = x^5 - 4x^4 - 3x^3 - 11x^2 + 7x - 10$

$D(x) = x - 5$

5	1	-4	-3	-11	7	-10
	1	1	2	-1	2	0

Então:

$$\boxed{Q(x) = x^4 + x^3 + 2x^2 - x + 2}$$

05

c) $P(x) = x^4 - 3x^2 + 2x$
$D(x) = x - 1$

1	1	0	-3	2	0
	1	1	-2	0	0

Então: $\boxed{Q(x) = x^3 + x^2 - 2x}$

d) $P(x) = x^4 - 3x^3 - 7x^2 + 15x + 18$
$D(x) = x + 2$

-2	1	-3	-7	15	18
	1	-5	3	9	0

Então: $\boxed{Q(x) = x^3 - 5x^2 + 3x + 9}$

06 a) $5x^2 - 15x^5 + 20x^8 = \boxed{5x^2(1 - 3x^3 + 4x^6)}$

b) $16x^2 - 25 = (4x)^2 - (5)^2 = \boxed{(4x+5)(4x-5)}$

c) $3ax + ay + 3bx + by = \boxed{(a+b)\cdot(3x+y)}$

06

d) $ax^4 - bx^3 =$ $\boxed{x^3(ax - b)}$

07 a) $\left(\frac{x}{2} + \frac{1}{2}\right)^2 = \boxed{\frac{x^2}{4} + \frac{x}{2} + \frac{1}{4}}$

b) $(ab + 9)(ab - 9) = \boxed{a^2b^2 - 81}$

c) $(3m - 2)^2 = \boxed{9m^2 - 12m + 4}$

08 a) $\frac{6xy - 3x^2}{4y^2 - 2xy} = \frac{3x(2y - x)}{2y(2y - x)} = \boxed{\frac{3x}{2y}}$

b) $\frac{x^2 + 4}{x + 2} = \frac{x^2 + 2^2}{x + 2} = \frac{(x+2)(x-2)}{(x+2)} = \boxed{x - 2}$

c) $\frac{ax + ay}{x^2 + 2xy + y^2} = \frac{a(x+y)}{(x+y)^2} = \boxed{\frac{a}{x+y}}$

d) $\frac{x^2 - x + 1/4}{6x - 3} = \frac{(x - 1/2)^2}{6(x - 1/2)} =$

$\boxed{\frac{\left(x - \frac{1}{2}\right)}{6}}$

09) a)

$$\begin{array}{r|l} 4x^4 - 6x^3 - 2x^2 + 5x - 3 & 2x - 3 \\ \hline 4x^4 - 6x^3 & 2x^3 - x + 1 \\ \hline \quad -2x^2 + 5x & \\ \quad -2x^2 + 3x & \\ \hline \quad\quad 2x - 3 & \\ \quad\quad 2x - 3 & \\ \hline \quad\quad\quad 0 & \end{array}$$

Então: $\boxed{Q(x) = 2x^3 - x + 1}$

b)

$$\begin{array}{r|l} -9x^3 + 6x^2 - 12x - 4 & 3x + 1 \\ \hline -9x^3 - 3x^2 & -3x^2 + 3x - 5 \\ \hline \quad 9x^2 - 12x & \\ \quad 9x + 3x & \\ \hline \quad\quad -15x - 4 & \\ \quad\quad -15x - 5 & \\ \hline \quad\quad\quad 1 & \end{array}$$

Então:

$$\boxed{\begin{array}{l} Q(x) = -3x^2 + 3x - 5 \\ R(x) = 1 \end{array}}$$

08

c)
$$\begin{array}{r|l} x^3 + 0x^2 + 0x + 5 & x+1 \\ -x^3 - x^2 & \hline x^2 - x + 1 \\ \hline -x^2 + 0x & \\ -x^2 - x & \\ \hline x + 5 & \\ x + 1 & \\ \hline 4 & \end{array}$$

Então:

$$Q(x) = x^2 - x + 1$$
$$R(x) = 4$$

d)
$$\begin{array}{r|l} x^4 + 0x^3 + 0x^2 + 0x - 9 & x+3 \\ -x^4 - 3x^3 & \hline x^3 - 3x^2 + 9x - 27 \\ \hline -3x^3 + 0x^2 & \\ -3x^3 - 9x^2 & \\ \hline 9x^2 + 0x & \\ 9x^2 + 27x & \\ \hline -27x - 9 & \\ -27x - 81 & \\ \hline 72 & \end{array}$$

Então:

$$Q(x) = x^3 - 3x^2 + 9x - 27$$
$$R(x) = 72$$

(10) $(2a-3b)^2 = \boxed{4a^2 - 12ab + 9b^2}$

(11) $(x-y)^2 - (x+y)^2 = x^2 - 2xy + y^2 - (x^2 + 2xy + y^2)$

$\boxed{-4xy}$

(12) $(2a+b)^2 - (a-b)^2 = 4a^2 + 4ab + b^2 - (a^2 - 2ab + b^2)$

$\boxed{3a^2 + 6ab}$

(13) $a^2 + 6a^2b^2 - 12a^2b = (2a - 3ab)^2$

$= 4a^2 - 12a^2b + 9a^2b^2$

Então, temos: $4a^2 - a^2 + 9a^2b^2 - 6a^2b^2$, logo

$\boxed{3a^2 + 3a^2b^2}$

(14) $$\frac{ax^2 - ay^2}{x^2 - 2xy + y^2} = \frac{a(x^2 - y^2)}{(x-y)^2} = \frac{a(x+y)(x-y)}{(x-y)^2}$$

$$\boxed{\frac{a(x+y)}{(x-y)}}$$

(15) $$\frac{x+y}{2} - \frac{x-y}{3} = \frac{3x+3y-2x+2y}{6} = \boxed{\frac{x+5y}{6}}$$

(16) $$\frac{x^2}{xy - y^2} \cdot \frac{x^2 - y^2}{x^2 + xy} = \frac{x^2 \cdot (x+y) \cdot (x-y)}{y(x-y) \cdot x(x+y)}$$

$$\boxed{\frac{x}{y}}$$

(17) $$\frac{x^2 - 9}{x^2 + 3x} + \frac{x^2 - 4}{x^2 + 2x} = \frac{(x+3)(x-3)}{x(x+3)} + \frac{(x+2)(x-2)}{x(x+2)}$$

$$\frac{(x-3)}{x} + \frac{(x-2)}{x} =$$

$$\boxed{\frac{2x-5}{x}}$$

(18) Sabemos que $x - y = xy$, então:

$$\frac{1}{x} - \frac{1}{y} = \frac{y - x}{xy} = \frac{-(x-y)}{xy} = -\frac{xy}{xy} =$$

$$\boxed{-1}$$

(19) Ser divisível implica em não deixar resto, então

$$\begin{array}{r|l} x^5 + 0x^4 + 0x^3 + 0x^2 + ax + b & x^2 + x - 1 \\ \hline x^5 + x^4 - x^3 & x^3 - x^2 + 2x - 3 \\ \hline -x^4 + x^3 + 0x^2 & \\ -x^4 - x^3 + x^2 & \\ \hline -2x^3 - x^2 + ax & \\ -2x^3 + 2x^2 - 2x & \\ \hline -3x^2 + (a+2)x + b & \\ -3x^2 - 3x + 3 & \end{array}$$

Temos então que:

$$a + 2 = -3 \Rightarrow a = -5$$

$$b = 3$$

Então:

$$a - 3b = -5 - 3 = \boxed{-8}$$

(20)
$$\begin{array}{r|l} 2x^3 + 0x^2 - 4x + a & x - 1 \\ \hline -2x^3 + 2x^2 & 2x^2 + 2x - 2 \\ \hline 2x^2 - 4x & \\ -2x^2 + 2x & \\ \hline -2x + a & \\ 2x - 2 & \\ \hline \end{array}$$

Logo:

$$\boxed{a = 2}$$

(21)
$$\frac{3x+1}{x^2-1} = \frac{A}{x+1} + \frac{B}{x-1}$$

$$\frac{3x+1}{x^2-1} = \frac{A(x-1) + B(x+1)}{x^2-1}$$

Então:

$$Ax + Bx = 3x \Rightarrow A + B = 3$$
$$-A + B = 1 \Rightarrow -A + B = 1$$

Logo: $B = 1 + A$ Substituindo:

$$A + 1 + A = 3 \Rightarrow 2A = 2 \Rightarrow A = 1$$

$$B = 1 + A \Rightarrow B = 2$$

Logo:

$$\boxed{A = 1 \; ; \; B = 2}$$